Digital Electronics (NECE 102)

(Digital Counter)

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Counters in Digital Logic

- ✓ Digital Counters are a specific type of sequential logic circuit.
- ✓ Used to count and store the number of occurrences of events or processes.
- ✓ Used for the frequency divider
- ✓ Counters are made using flip-flops to store count values and respond to clock signals

Digital Counters - Key Characteristics

- **Maximum Count:** Determined by the number of flip-flops (2^n states for n flip-flops)
- **Up/Down Counting:** Ability to count in ascending or descending order
- **Synchronous/Asynchronous Operation:** Timing of state changes
- **Free-Running/Self-Stopping:** Continuous or limited counting
- **Positive or Negative edge triggered:** depending on the connection provided at the clock input of the flip-flops.

The behavior of a counter can be explained in terms of timing diagram (waveforms), truth table and/or state diagrams.

Types of Counters

- Asynchronous (Ripple) Counters
- Synchronous Counters
- Up/Down Counters

Advanced Counter Concepts

- Decade Counters: Count from 0 to 9 in decimal
- Ring Counters: Circular shift register-based counters
- Johnson Counters: Twisted ring counters
- Gray-Code Counters: Output Gray code sequences

Counter Design Considerations

- Modulus:
  - The *modulus* of a counter is the number of states in its count sequence.
  - A counter that has modulus value *n* is commonly referred to as a *modulo-n* or *MOD-n* counter. For example, a counter that counts up from 0 to 5 and then overflows is a MOD-6 counter because it has six states.
  - The maximum possible modulus of a counter is determined by the number of flip-flops

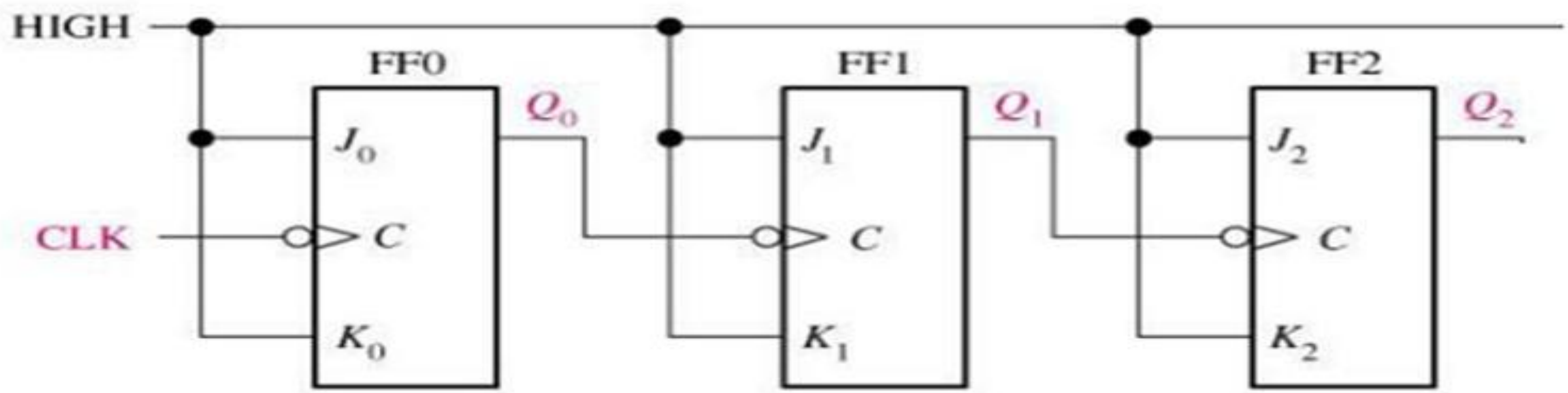
- Number of Flip-Flops: Determined by desired maximum count (2^n)

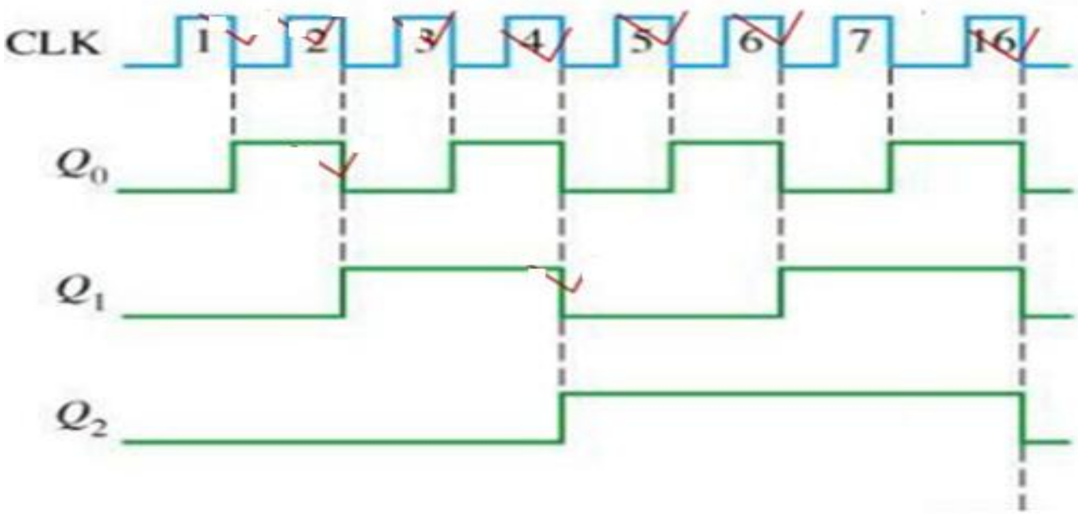
- Edge Triggering: Positive or negative edge-triggered designs

Asynchronous (Ripple) Counters

- Clock pulses "ripple" through flip-flops sequentially
- Simpler design but slower operation due to propagation delays

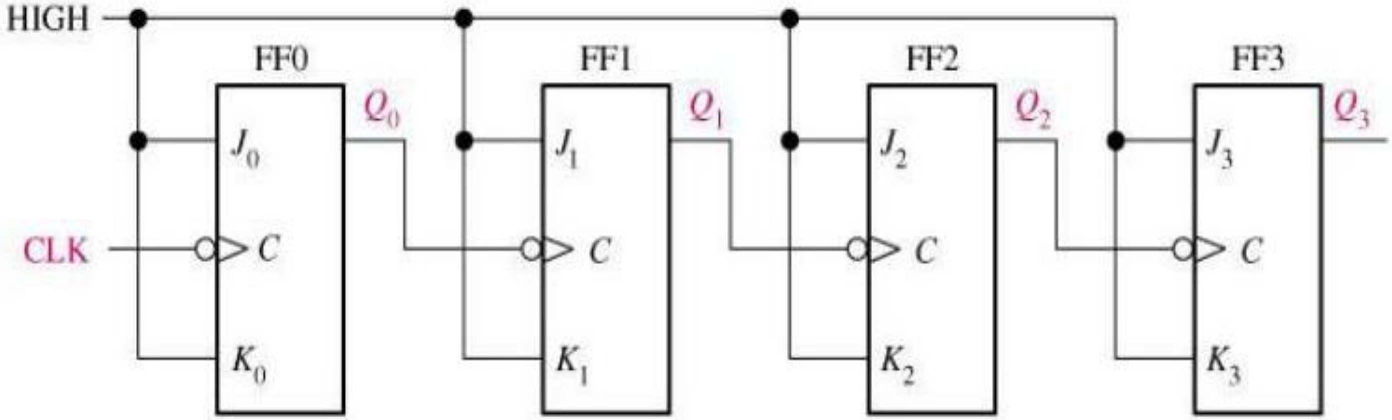
Asynchronous (Ripple) Counters

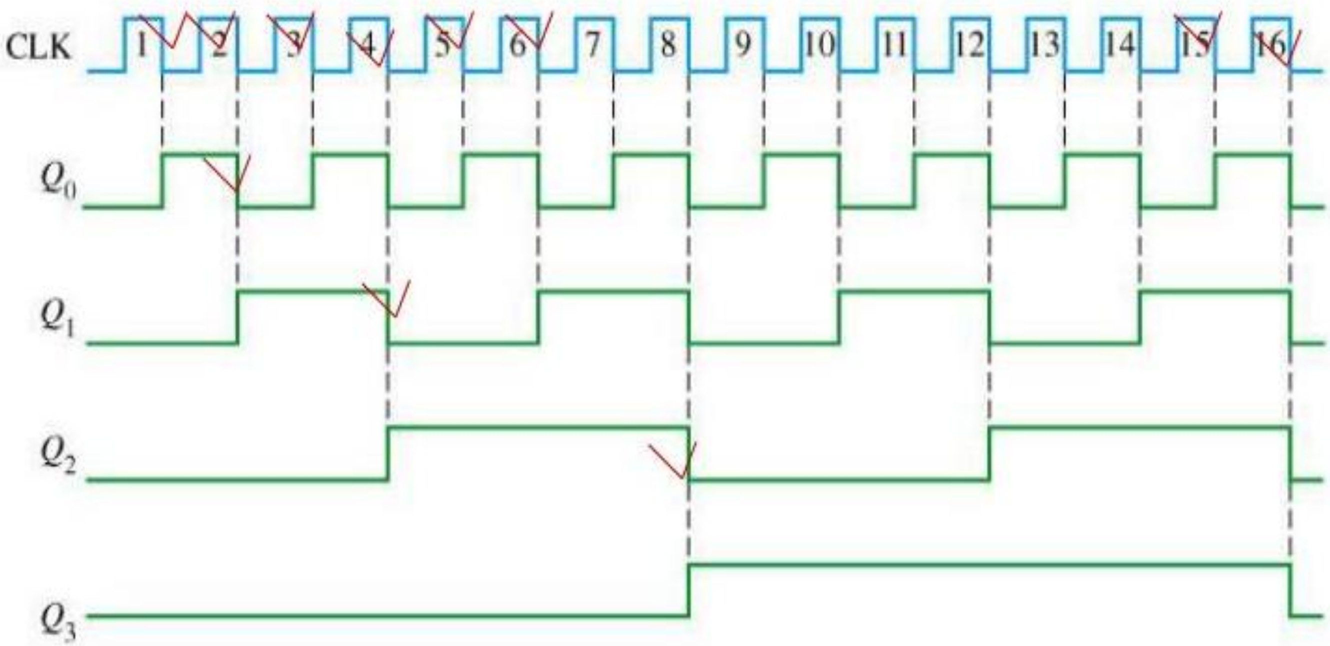




CLK PLUSE	Q2	Q1	Q0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
REPEAT	0	0	0

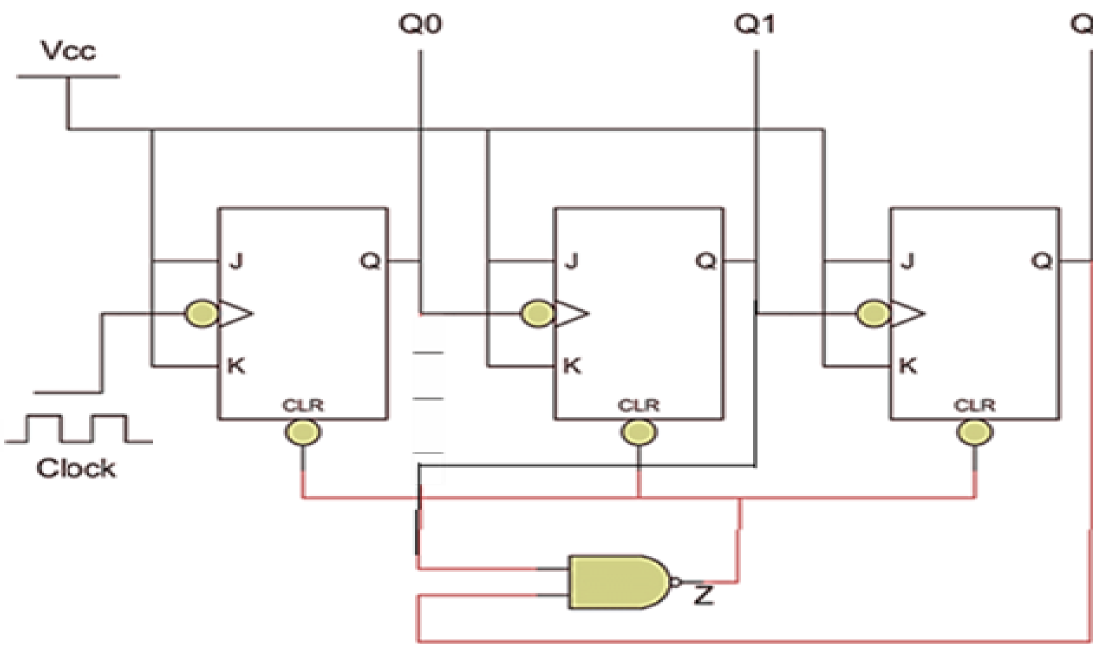
4 –bit Asynchronous (Ripple) Counters

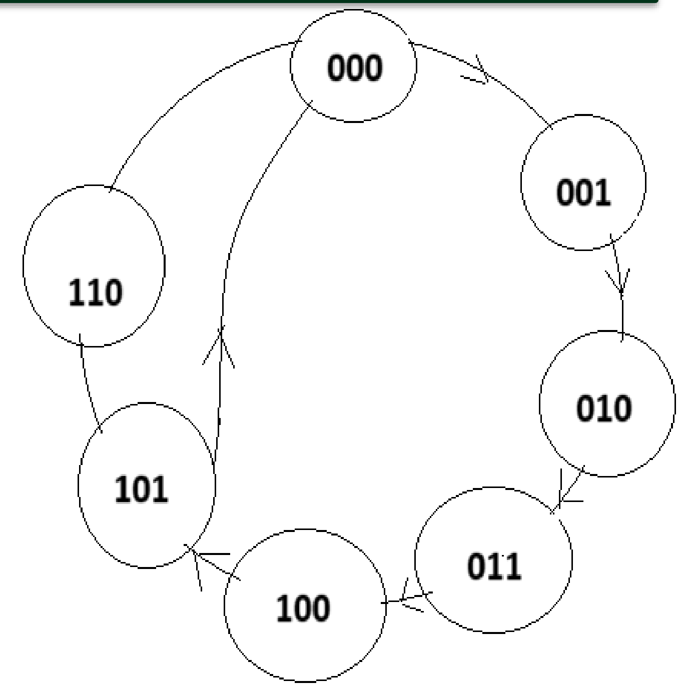


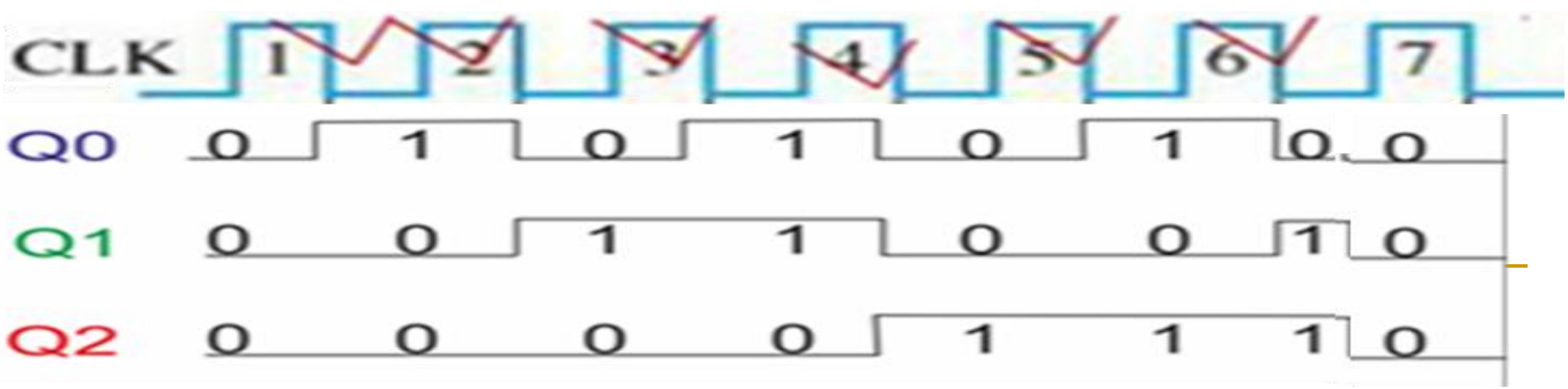


CLK PLUSE	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16 REPEAT	0	0	0	0

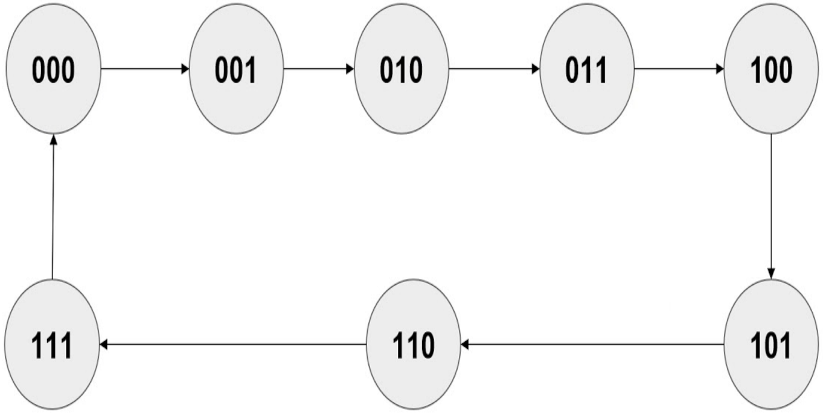
MOD 6 Counter (Asynchronous)

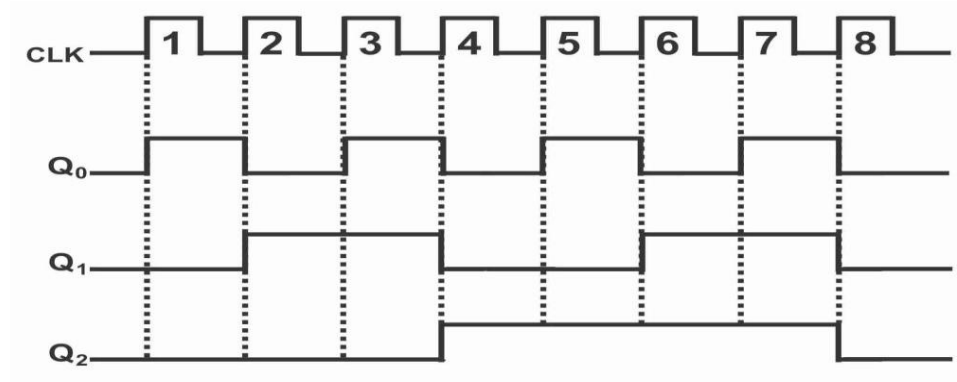






3 bit synchronous counter





Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	1	0
1	0	1

Q_2	Q_1	Q_0	Q_2^*	Q_1^*	Q_0^*	T_2	T_1	T_0
0	0	0	0	0	1			
0	0	1	0	1	0			
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

3 bit synchronous counter

Q_2	Q_1	Q_0	Q_2^*	Q_1^*	Q_0^*	T_2	T_1	T_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

$T_2 = Q_1 Q_0$

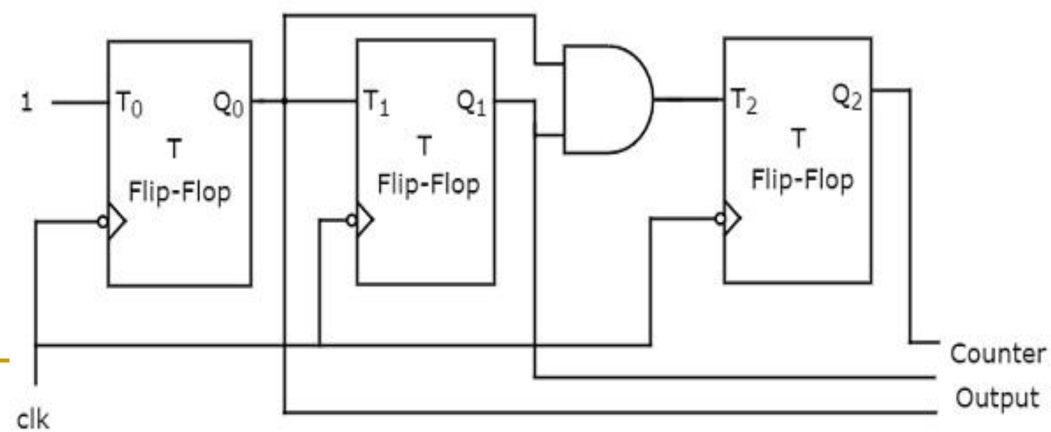
$\bar{Q}_2$ \ $Q_1 Q_0$	00	01	11	10
0	0	0	1	0
1	0	0	1	0

$T_1 = Q_0$

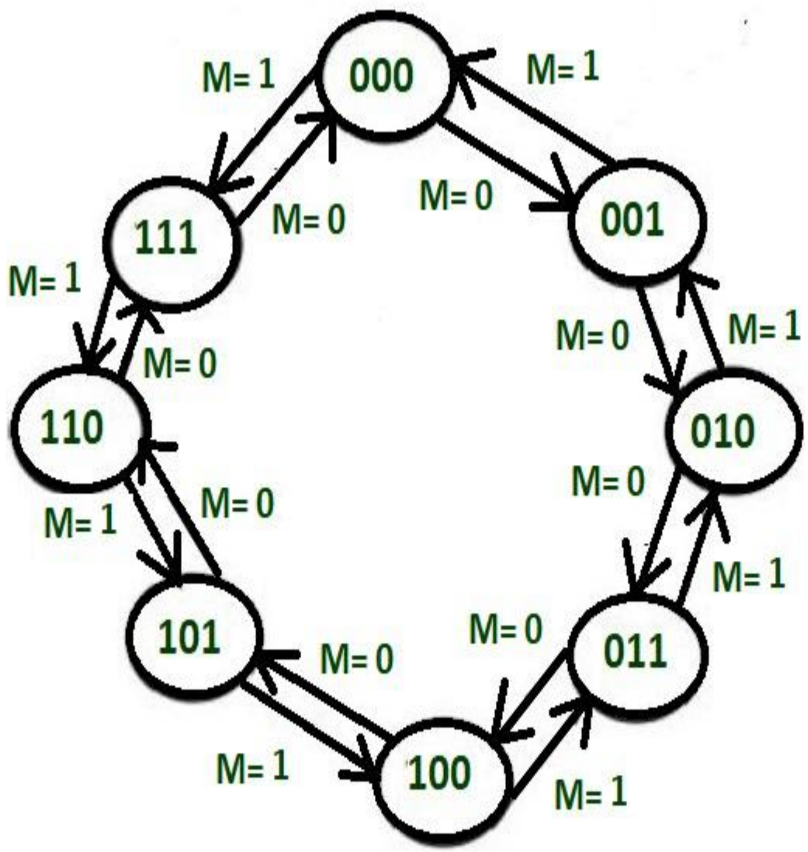
$\bar{Q}_2$ \ $Q_1 Q_0$	00	01	11	10
0	0	1	1	0
1	0	1	1	0

$T_0 = 1$

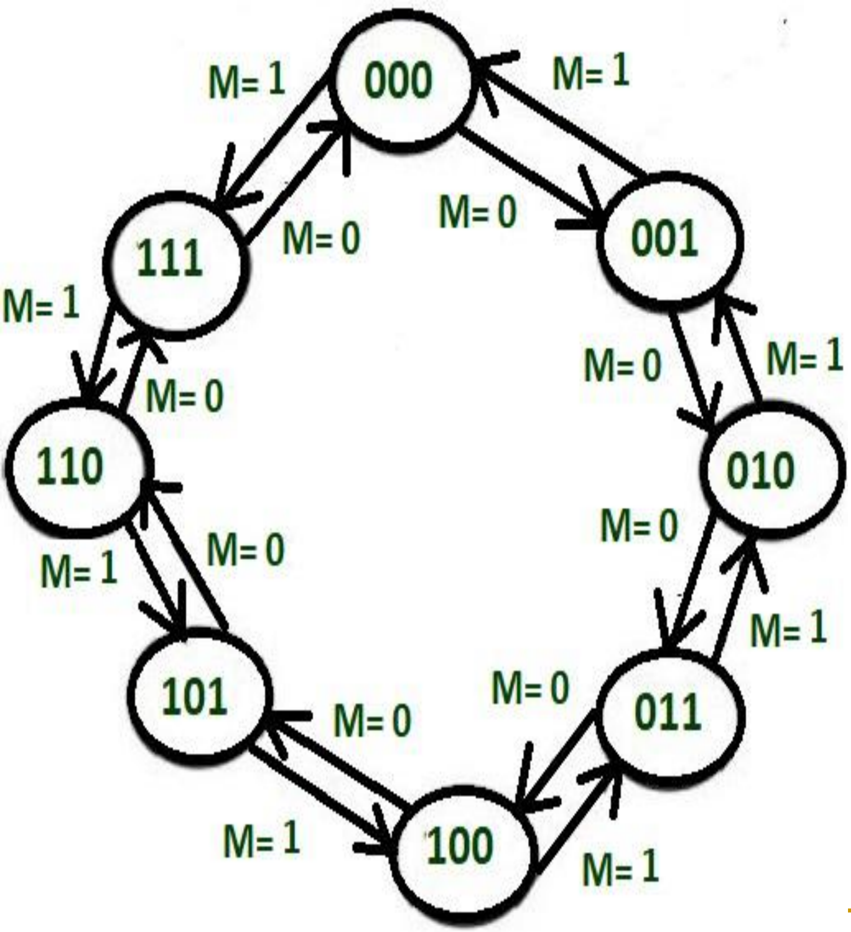
$\bar{Q}_2$ \ $Q_1 Q_0$	00	01	11	10
0	1	1	1	1
1	1	1	1	1



3 bit synchronous up/Down Counter



3 bit synchronous up/Down Counter



M	Q_3	Q_2	Q_1	Q_3^*	Q_2^*	Q_1^*	T_3	T_2	T_1
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

3 bit synchronous up/Down Counter

$MQ_3 \backslash Q_2Q_1$	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	0	0	0
10	1	0	0	0

$$\boxed{T_3 = M'Q_2Q_1 + MQ'_2Q'_1}$$

$MQ_3 \backslash Q_2Q_1$	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$$\boxed{T_2 = M'Q_1 + MQ'_1}$$

$MQ_3 \backslash Q_2Q_1$	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$\boxed{T_1 = 1}$$

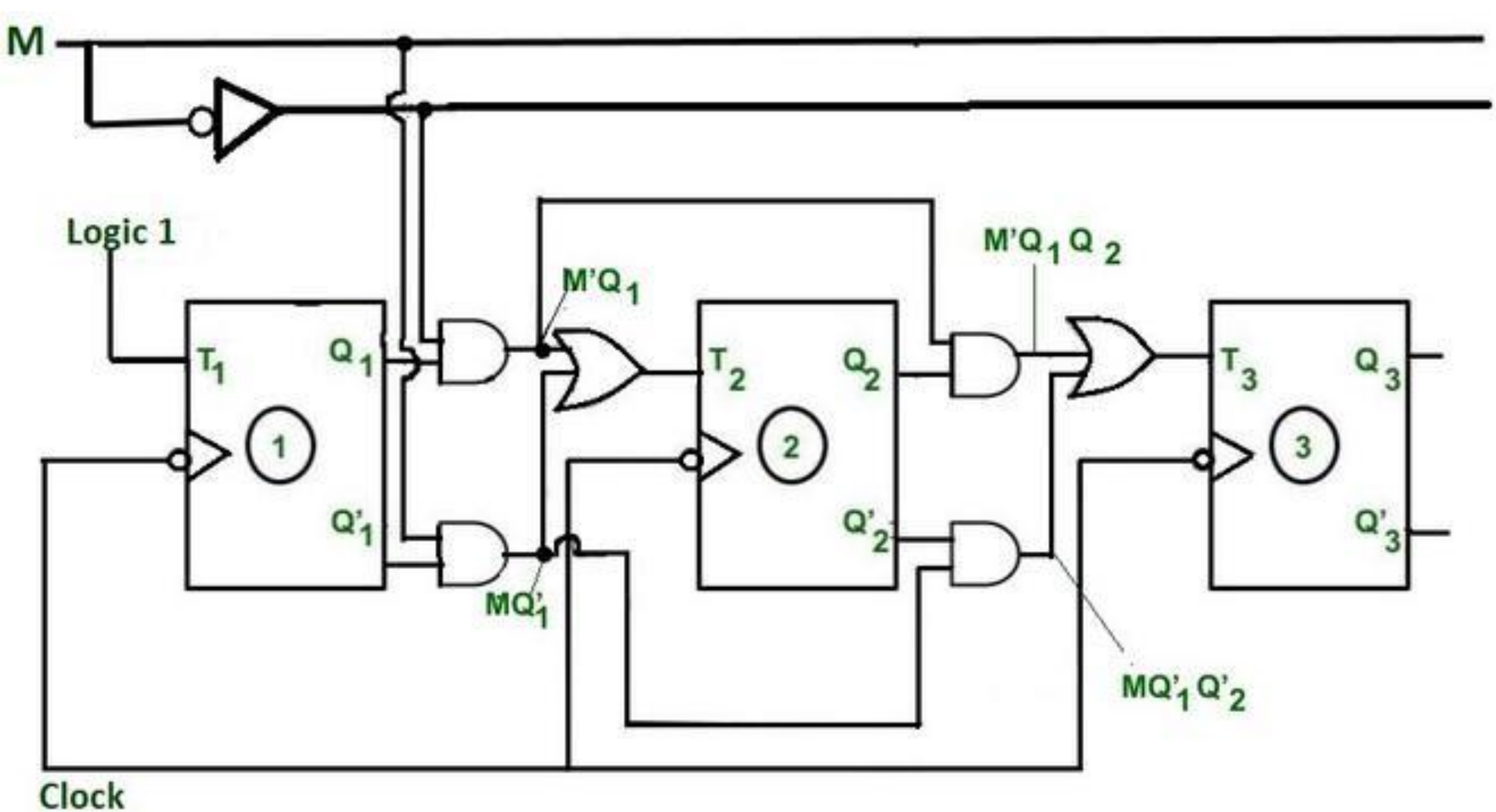
M	Q_3	Q_2	Q_1	Q_3^*	Q_2^*	Q_1^*	T_3	T_2	T_1
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

3 bit synchronous up/Down counter

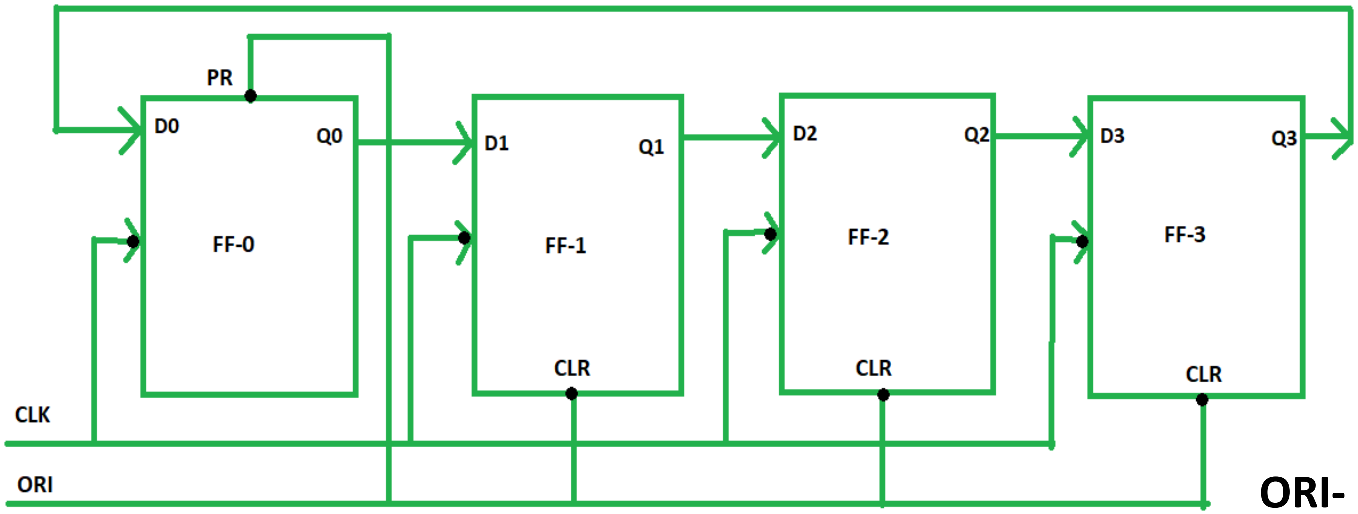
$T_3 = M'Q_2Q_1 + MQ'_2\,Q'_1$

$T_2 = M'Q_1 + MQ'_1$

$T_1 = 1$



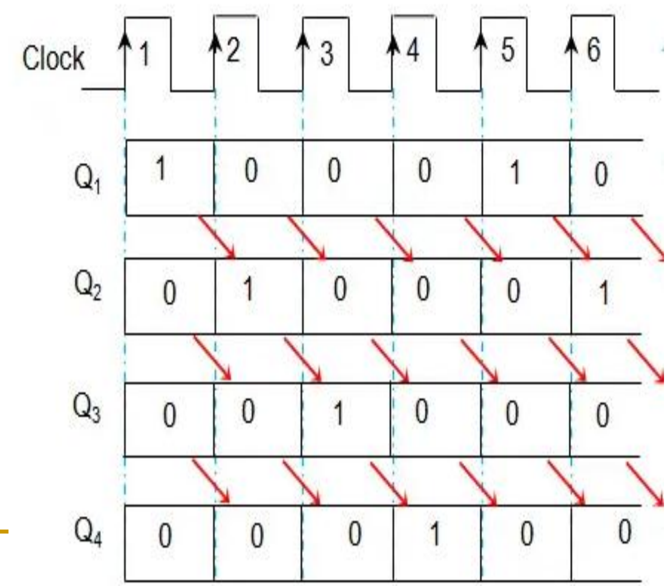
Ring counter



ORI- Over riding input

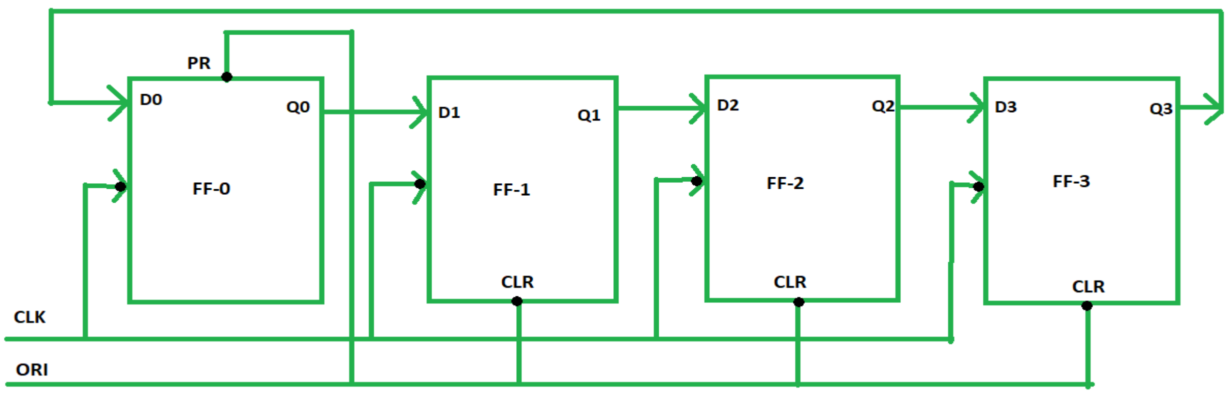
Ring Counter

ORI	Clk	Q_0	Q_1	Q_2	Q_3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0



Output waveform of 4-bit ring counter

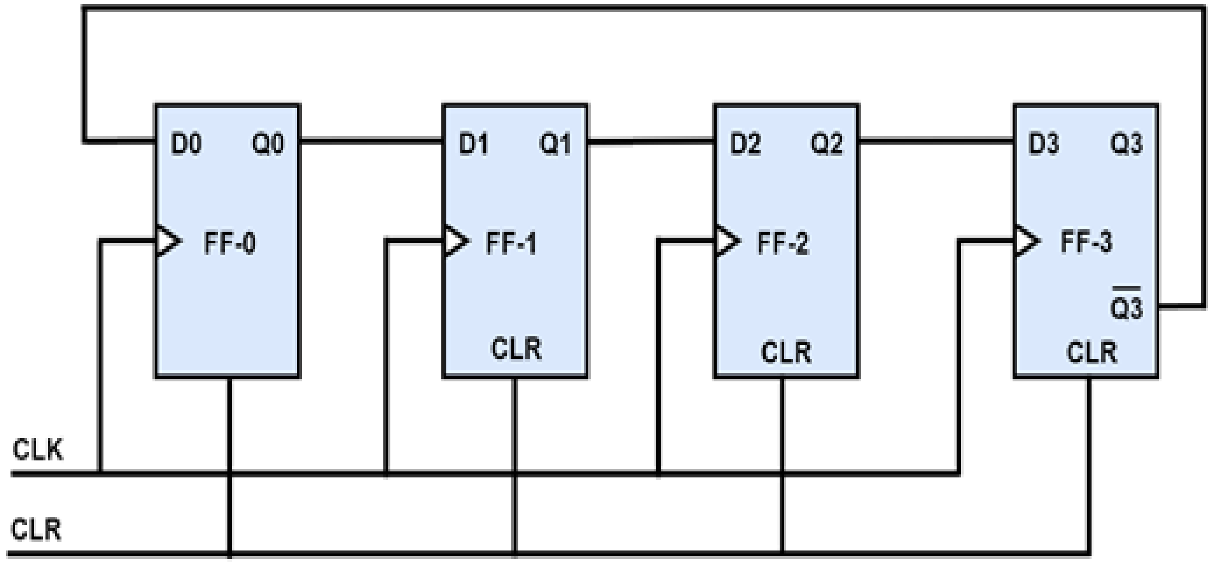
Ring counter



Ring Counter

ORI	Clk	Q_0	Q_1	Q_2	Q_3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

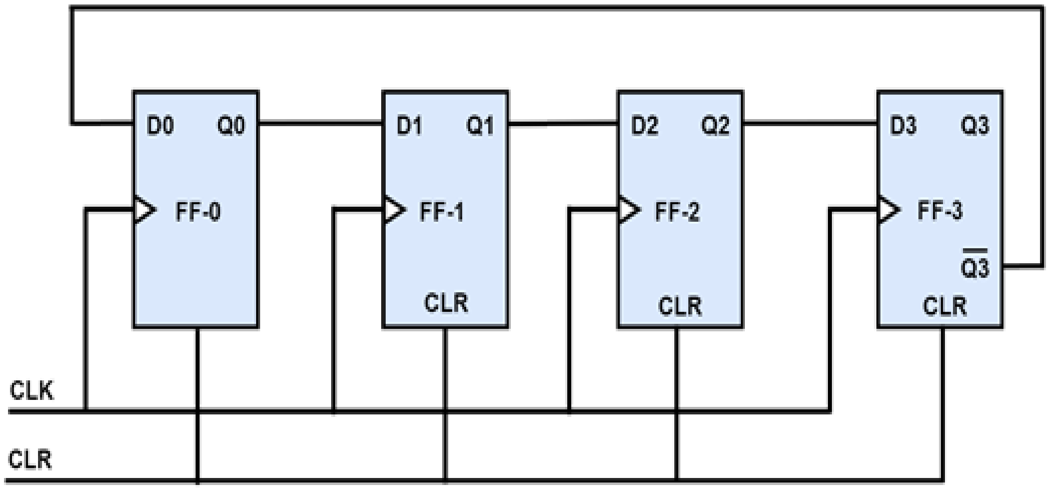
Johnson Counter



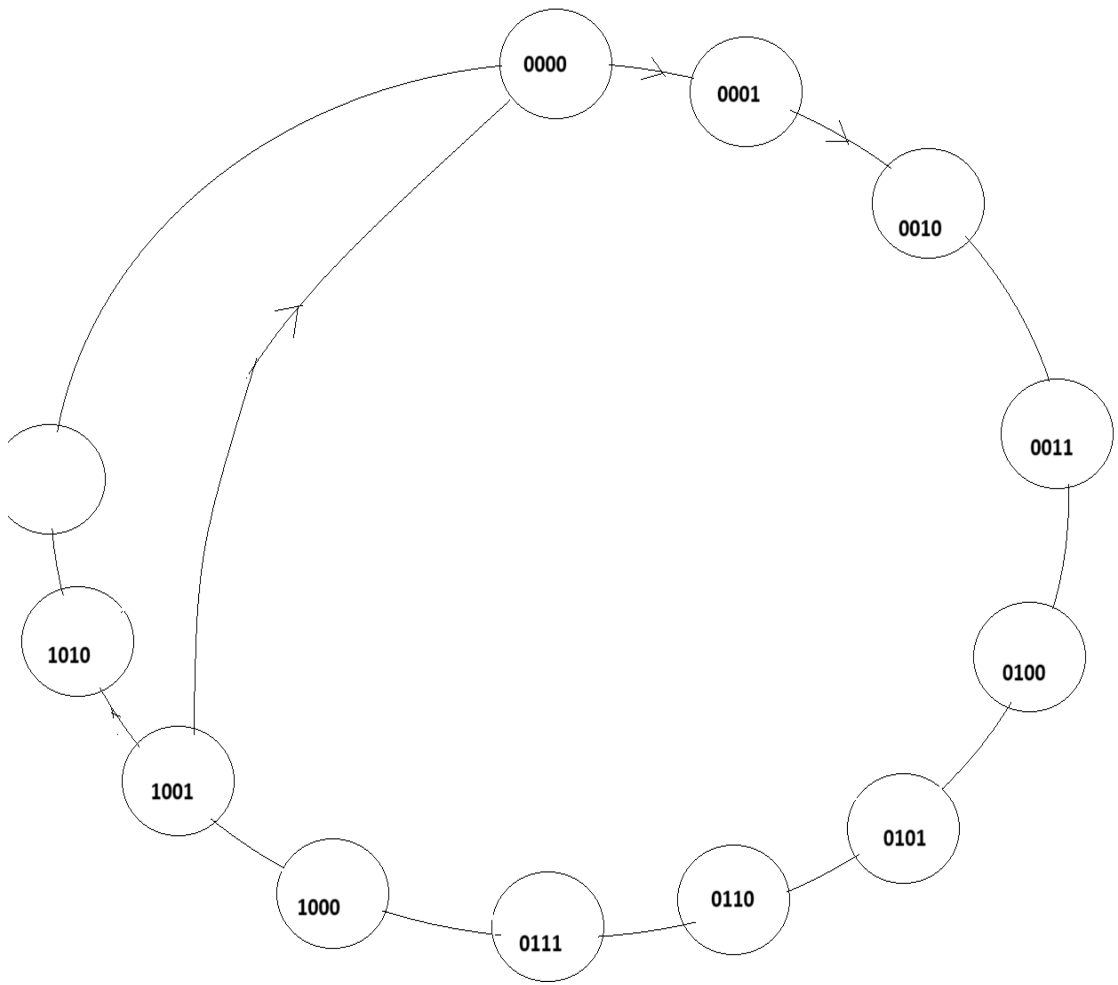
CP	Q1	Q2	Q3	Q4
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

Johnson Counter

CP	Q1	Q2	Q3	Q4
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

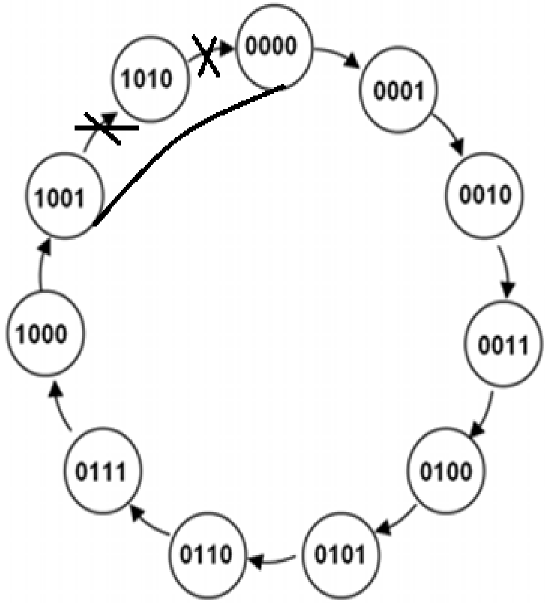


4 bit synchronous Decade Counters



4 bit synchronous Decade Counters

State diagram and State table of decade counter



Present State				Next state							
Q_3	Q_2	Q_1	Q_0	Q_3'	Q_2'	Q_1'	Q_0'	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	1	0
1	0	1

4 bit synchronous Decade Counters – Input output relationship (using K Map

Present State				Flip Flop Input			
Q_3	Q_2	Q_1	Q_0	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	1
0	0	1	0	0	0	0	1
0	0	1	1	0	1	1	1
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1
0	1	1	0	0	0	0	1
0	1	1	1	1	1	1	1
1	0	0	0	0	0	0	1
1	0	0	1	1	0	0	1

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	X	X	X	X
10	0	1	X	X

$$\boxed{T_3 = Q_3Q_0 + Q_2Q_1Q_0}$$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	X	X	X	X
10	0	0	X	X

$$\boxed{T_2 = Q_1Q_0}$$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	X	X	X	X
10	0	0	X	X

$$\boxed{T_1 = Q_3'Q_0}$$

M Q_3 \ Q_2Q_1	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$\boxed{T_0 = 1}$$

4 bit synchronous Decade Counters

$T_3 = Q_3Q_0 + Q_2Q_1Q_0$

$T_2 = Q_1Q_0$

$T_1 = Q'_3Q_0$

$T_0 = 1$